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Perspective

How does cytoplasmic crowding affect reaction rates?

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SUMMARY

The cytoplasm of eukaryotic cells is crowded with macromolecules. In principle, this crowding could have either a positive or a negative effect on the rates of biochemical reactions. Here, we review two commonly invoked theories to account for these possible effects then survey recent experimental work in cells and extracts that measures the effects. The evidence so far suggests that the effective second-order rate constants (a measure of the speed of a reaction for a given concentration of reactants) for reactions *in vivo* generally go down when crowding increases due to the slowing of diffusion. If the evidence presented so far proves to be general, it would have important implications for how we view the trade-offs that determine the biochemical dynamics of the cytoplasm.

INTRODUCTION

It has long been recognized that biochemistry operates differently in living cells than in test tubes.[1,2] Differences between living cytoplasm and most *in vitro* systems include molecular crowding,[3,4] soft interactions,[5,6] the fractal-like geometry of the cell interior,[7,8] the colloid-like[9–12] or gel-like[13,14] properties of the protoplasm, and the capacity to self-organize into large-scale patterns[15–19] and movements.[20–24] Here, we focus on crowding and how it affects the speeds of cellular biochemical reactions, because this is arguably the most fundamental and easy-to-understand difference between living systems and *in vitro* systems. For reactions, we will examine several enzyme-mediated catalytic processes that can be analyzed via two-step Michaelis-Menten models. However, a similar analysis could be applied to binding reactions, condensation reactions, or protein folding, among others.

We begin with an overview of how crowded cells are, then review two frequently invoked theories for the consequences of crowding: one describing how crowding could negatively impact reaction rate constants and one for how it could positively affect them. A deeper dive into the historical development of the concept of crowding and additional theoretical approaches can be found in Alfano et al.[25] We then compare these theories to recent experimental results from yeast cells and *Xenopus laevis* egg extracts that have been manipulated to be more or less crowded than normal. So far, the results seem quite surprising.

How crowded are cells?

The crowders in a eukaryotic cell include proteins, nucleic acids, polysaccharides, and organelles. *A priori* it seems like organelles might be the most important cytoplasmic crowders, given that they may occupy as much as half the cytoplasmic volume.[26] However, at least some organelles, like the nucleus, are so large and static that they probably have little effect on most cytoplasmic reactions (unless the reactants are very close to the nucleus). In addition, the diffusion of protein-sized probes as assessed by fluorescence correlation spectroscopy (FCS, which assesses diffusion over distance scales of ~100–200 nm, which is comparable to the distance between a protein and an average-abundance 200 nM target) is very similar in organelle-free cytosolic *Xenopus* egg extracts and self-organized, organelle-containing *Xenopus* extracts,[27] suggesting that organelles do not have much of an effect on the dynamics of these probes. Organelles may well affect the dynamics of some species over some timescales and distance scales, but here, we will focus on macromolecules as the principal cytoplasmic crowders.

Proteins are the most abundant of the cytosolic macromolecules, and in cultured mammalian cells, the typical protein concentration is ~85 mg mL^{-1}.[28] Assuming an average protein weight of ~45 kDa, the molar concentration of proteins is ~2 mM. For a colloidal suspension at 2 mM concentration, the average center-to-center distance of the particles is ~10 nm.[29,30] If we assume proteins have an average mass density of 1.37 g mL^{-1} and are spherical in shape, then the average protein radius will be ~2.4 nm, making the surface-to-surface distance ~5 nm, or about the diameter of an average protein.[31,32] This is comparable to standing in a crowd of people where everyone is only a shoulder-width apart from each other. The volume occupied by the proteins is about 6% of the total cytoplasmic volume,[28–34] a number that might seem surprisingly low given the images emerging from cryo-electron tomography (cryo-ET) reconstructions of the cytoplasm[35,36] as well as Goodsell's beautiful watercolors of cellular structures (https://pdb101.rcsb.org/sci-art/goodsell-gallery). Other macromolecules, especially RNAs, occupy a substantial fraction of the cytosolic volume as well, yielding a total occupied fraction of perhaps 10%. By this measure, the volume fraction of the cytoplasm that is inaccessible to small molecule

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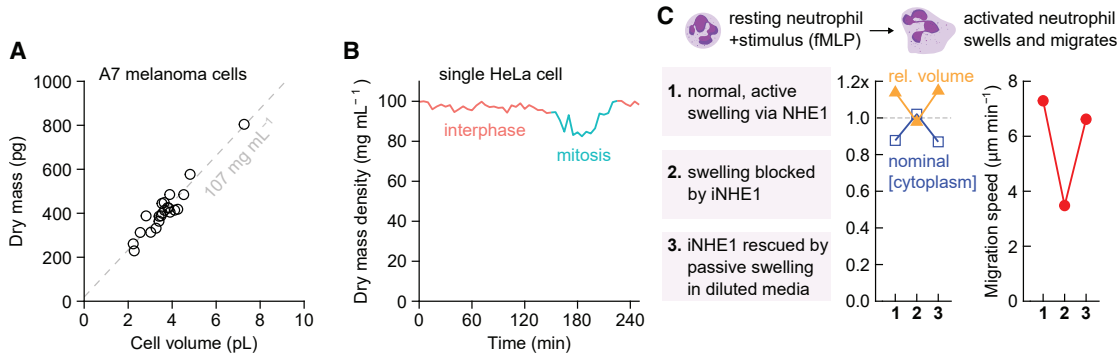

Figure 1. Precise maintenance and functional regulation of cellular crowding

(A) Dry mass (protein and lipid mass) and volume measured in 22 cells of a human A7 melanoma cell line by normalized Raman imaging.[28] Each point represents a single cell. Their volumes and dry masses are distributed over a ~4-fold range. A straight line could be fitted to all 22 cells, with a slope of 107 mg mL^{-1} and a *y*-intercept at 19.4 pg with R^2 of 0.92.

(B) Dry mass density of a single HeLa cell measured by quantitative phase imaging for dry mass and fluorescence exclusion microscopy for cell volume.[53] The cell swelled during mitosis and diluted the dry mass. Dry mass density was stable during interphase (here showing ~3 h outside of mitosis) with minor fluctuations.

(C) Migration speed and cell volume measurements for neutrophils after 25 min of treatment with N-formyl-methionyl-leucyl-phenylalanine (fMLP).[54] The stimulated neutrophils actively swelled through the agency of the NHE1 Na^+/H^+ exchanger to ~115% of their original volume and increased migration speed to 7.3 μm min^{-1}. Treatment with the NHE1 inhibitor (iNHE1) prevented swelling and decreased migration speed to 3.5 μm min^{-1}. Rescuing iNHE1 with 20% diluted media restored swelling to ~115% volume and migration speed to 6.6 μm min^{-1}. Thus, a 15% swelling and ~13% dilution of the dry mass or cytoplasmic concentration strikingly led to a ~100% increase in migration speed.

probes, denoted by ϕ, would equal ~0.1. The remaining ~90% of the volume is water.[37] If some of the macromolecules are present in relatively high-density condensates, the value of ϕ for the non-condensed part of the cytoplasm could be lower.

The volume fraction ϕ can alternatively be estimated from measurements of cell volume vs. extracellular osmolarity, using the Boyle van't Hoff equation and Ponder plots to calculate the osmotically inactive volume of the cell.[38–40] Typical reported values are $\phi \approx 0.25$, or 25% of the total cell volume.[9,41–45] The discrepancy between these estimates and the smaller values obtained from the measured mass and density of cellular macromolecules could mean that ~15% of the cell volume is inexchangeable water, presumably tied up with macromolecules on the timescale of the experiments performed.[46] Note though that the extent to which cellular water is bound to macromolecules is a matter of debate, with estimates ranging from nearly 100% bound[47,48] to nearly 100% freely exchangeable.[32] Based on the direct measurements of mass density and the inferred estimate of the osmotically inactive volume fraction (0.25), here we will assume that the fraction of a cell occupied by macromolecules plus inexchangeable water is between 0.1 and 0.25.

Precise maintenance of macromolecular concentration

Within a particular cell type, the cell-to-cell variation in the protein concentration is quite small, with reported coefficients of variation of about 9%.[28] By contrast, the variation in cell volume and mass is much greater, typically on the order of 20%–30%.[28] Cell mass and volume do not vary independently, and the overall macromolecular concentration is fairly precisely maintained (Figure 1A). This near-constancy of macromolecular concentration suggests that the number of macromolecules in a cell determines the cell volume, or that the cell volume determines the number of macromolecules, or both. Untangling how cell volume and mass are normally so tightly coupled remains an important challenge for the field.[49–52]

Nevertheless, in different cell types and tissues, the concentration of macromolecules does vary quite substantially.[37] In some specialized cell types, such as red blood cells, protein concentration is on the order of ~330 mg mL^{-1}, 3 to 4 times as high as the typical concentrations in mammalian cell lines.[28,31,32,37] Different types of brain cells vary in protein concentration, with high-density neurons standing out from a background of lower-density cells in brain slices.[37] When mouse embryonic stem cells differentiate into the neuronal lineage, their cytoplasm becomes ~10% more dilute (shifting from 140 to 125 mg mL^{-1}), and this may explain differences in spindle size scaling after differentiation.[55] Growth plate-residing chondrocytes become ~55% more dilute (shifting from 180 to 80 mg mL^{-1}) as they undergo terminal differentiation and move closer to the cartilage-bone interface.[56] Even within a particular cell type, changes in macromolecular concentration occur. For example, mitotic cells are ~15% larger and lower in density than interphase cells (Figure 1B).[37,57,53] Neutrophils actively swell and dilute their cytoplasm by ~13% when activated, allowing them to migrate ~100% faster than the non-swollen cells (Figure 1C).[54] Likewise, lymphocytes reversibly change density upon activation or inactivation.[58] Even attachment to a solid surface[44] and cell migration can cause ~10% fluctuations due to mechano-osmotic coupling.[44,54] Changes in concentration also occur in pathological processes, such as cell senescence (up to 50% dilution)[28,37,59] and chondrocyte hypertrophy in osteoarthritis (~50% dilution),[60] or under environmental perturbations such as osmotic shock, mechanical pressure, and desiccation.[61–64] These observations suggest that the concentration of macromolecules, while actively maintained, is changed in specific biological contexts. This might be expected to broadly affect the speeds of biochemical processes, which brings us back to the question of how crowding might be *expected* to affect reaction rates, and how it *does* affect them.

Phillies's law and the negative effects of crowding on diffusion-controlled reactions

Whether an enzyme-mediated process is expected to be affected positively or negatively by crowding depends upon whether the process is diffusion controlled or reaction controlled. Theories that treat macromolecules as inert hard spheres predict that diffusion-controlled processes should be negatively affected and reaction-controlled processes positively affected.

To see this, we begin with the Michaelis-Menten model for an enzyme-catalyzed reaction, in which an enzyme (E) catalyzes the formation of product (P) from substrate (S):

$$E + S \underset{k_d}{\overset{k_a}{\rightleftharpoons}} ES \overset{k_{cat}}{\rightarrow} E + P \qquad \text{(Equation 1)}$$

Once the system has reached a steady-state concentration of enzyme-substrate complex (ES), the rate of the reaction is

$$rate = \frac{k_{cat}[E][S]}{K_M + [S]}, \qquad \text{(Equation 2)}$$

where the Michaelis constant $K_M = \frac{k_{cat}+k_d}{k_a}$. If the substrate concentration is low compared with the K_M value (i.e., the enzyme is far from saturation), the rate equation collapses to a mass action expression:

$$rate = \frac{k_{cat}}{K_M}[E][S] = \frac{k_{cat}k_a}{k_{cat}+k_d}[E][S]. \qquad \text{(Equation 3)}$$

We can therefore think of the quantity $\frac{k_{cat}k_a}{k_{cat}+k_d}$, termed $k_{effective}$, as the effective second-order rate constant for a far-from-saturation reaction:

$$k_{effective} = \frac{k_{cat}k_a}{k_{cat}+k_d}, \text{where } rate = k_{effective}[E][S]. \qquad \text{(Equation 4)}$$

Next consider two limiting cases. First, assume that the intermediate complex ES undergoes catalysis much more quickly than it undergoes dissociation back to E and S. That is, k_d is negligible compared with k_{cat}. This situation is referred to as the diffusion limit. In the diffusion limit, the enzyme is a perfect catalyst, and every collision leads to a reaction. It follows from Equation 4 that $k_{effective}$ equals k_a, and so the second-order rate constant in the diffusion limit is

$$k_{diffusion} = k_a. \qquad \text{(Equation 5)}$$

The association rate constant k_a is often taken to be the proportionality constant for the expression relating collision rate to the concentrations of the reactants. This collision rate, and its proportionality constant, are typically not measurable quantities, but they can be calculated theoretically. Assuming the enzyme and substrate are dilute hard spheres that neither attract nor repel the other, the theoretical collision rate is given by the Smoluchowski equation and is proportional to the sum of the enzyme's and substrate's diffusion coefficients.[65] Therefore:

$$k_{diffusion} = k_a \propto (D_E + D_S), \qquad \text{(Equation 6)}$$

where D_E is the diffusion coefficient of the enzyme, and D_S is the diffusion coefficient of the substrate. Crowders impede diffusion (Figure 2A), which lowers the enzyme-substrate collision rate and therefore slows down the overall rate of the reaction. Diffusion coefficients have been found to typically decrease approximately exponentially with increasing crowding, a relationship proposed by Minton[66] and Phillies.[67,68]

$$\frac{D}{D^0} = e^{-g\phi}, \qquad \text{(Equation 7)}$$

where D is the diffusion coefficient for some molecule in the crowded environment, D^0 is the diffusion coefficient in the uncrowded buffer, ϕ is the volume fraction of crowders, and g is a scaling factor. The exponential decrease of the diffusion coefficient is consistent with the exponential increase of viscosity in soft colloids approaching the glass transition.[9,75,76]

Figure 2B shows the diffusion coefficients for mesoscale polystyrene microspheres (where we take "mesoscale" to mean 10 to 1,000 nm in diameter[77,78]) in *Xenopus* egg extracts diluted to different concentrations and assessed by particle tracking.[69] As the extract is diluted, the diffusion coefficients increase, and, as predicted by Phillies's law, the diffusion coefficients decrease exponentially with increasing cytoplasmic concentration. Similar trends have been reported for the diffusion of GEMs, genetically encoded multimeric nanoparticles in budding yeast and fission yeast, although overall their diffusion coefficients are about an order of magnitude smaller than those of similar-sized microspheres in frog egg extracts.[45,79,80] Other things being equal, larger proteins have their diffusion coefficients, diffusion-controlled reaction rate constants ($k_{diffusion}$), and g values affected more strongly than smaller proteins. This pattern can be understood from the schematic view of a collision between a probe and a crowder shown in Figure 2A. A probe will collide with a crowder if its center comes to within a distance of $r_P + r_C$ of the center of the crowder, where r_P is the radius of the probe and r_C is the radius of the crowder (Figure 2A). Therefore, a crowder should collide more frequently with a big probe than with a small probe, and crowding should be more effective at slowing the diffusion of large species. Experiments in *Xenopus* extracts with mesoscale microspheres of varying size show this to be the case: the 40 nm beads are ~2.4-fold slower in 1× cytoplasm than in buffer, whereas the 200 nm beads are 29-fold slower (Figure 2B).[69,79]

Note that this size dependence becomes most apparent on the mesoscale. For small molecule probes and protein-sized probes (<40 nm), the experimentally determined ratio $\frac{D^{1x}}{D^0}$ is about 0.5, with a corresponding g value (assuming $\phi = 0.25$ as suggested by estimates of the osmotically inactive volume in typical cells) of about 3 and only a modest dependence on probe molecular weight (Figure 2C). However, the ratio $\frac{D^{1x}}{D^0}$ falls steeply with molecular weight for mesoscale probes, and the g values increase approximately linearly with increasing probe molecular weight (Figure 2C).[69,79,70,71]

If the diffusion g values for an enzyme and a substrate are approximately equal, the effective rate constant is

$$\frac{k_{diffusion}}{k^0_{diffusion}} = e^{-g\phi}. \qquad \text{(Equation 8)}$$

If the g values differ, the expression is more complicated, but in practice, a single exponential is often sufficient to account for experimental rate measurements (see Figure 4 below).

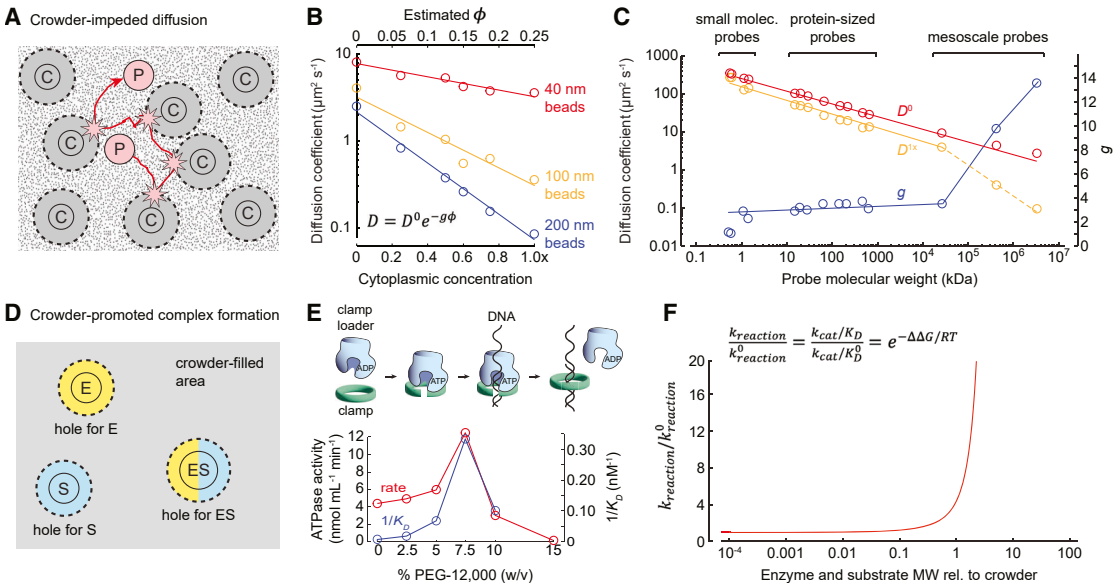

Figure 2. How crowding can both positively and negatively affect reaction rates

(A–C) Crowding can slow diffusion. (A) Schematic depiction of a probe molecule (designated P) colliding with crowders (designated C). Note that if the center of probe P comes within a distance of $r_p + r_c$ of the center of crowder C, P will bounce off C. Thus, if the probe is large, the excluded volume around C (bounded by the dashed circle) will extend well beyond the surface of C (solid circle). (B) The effect of cytoplasmic concentration on the diffusion of mesoscale microspheres in *Xenopus* egg extracts. Diffusion slows as the cytoplasmic concentration increases, and the effect of crowding is greater for larger probes than for smaller probes. Diffusion was measured by particle tracking, and the effective diffusion coefficients were calculated from the first 1 sec of data[69]. (C) The effect of cytoplasmic crowders on the diffusion of probes of various sizes in cultured human cells and *Xenopus* extracts.[69–71] The red circles represent data for the diffusion coefficients of probes in buffer (D^0). The fitted line represents the Stokes-Einstein relationship ($D \propto mw^{-1/3}$). The orange circles (D^{1x}) are data from astrocytes in culture (for the small molecules) or 1× *Xenopus* egg extracts (for the protein-sized molecules and mesoscale probes). Although the movement of macromolecules in cells is actually subdiffusive (i.e., the mean squared displacement is less than linearly proportional to time), over the time and distance scales relevant to the collision of an enzyme with a substrate, Brownian diffusion is generally a reasonable approximation. The corresponding calculated *g* values are shown in blue. Data for the small molecule and protein-sized probes were obtained by single-molecule displacement/diffusivity mapping and taken from Choi et al.[70,71] The data for mesoscale probes were obtained by particle tracking and taken from Chen et al.[69] Similar trends were reported by Seksek et al.[34] based on photobleaching experiments in MDCK and Swiss 3T3 cells microinjected with fluorescein isothiocyanate (FITC)-dextrans or FITC-Ficolls of various sizes, except that the observed ratio of D to D^0 was ~0.25 rather than the ~0.5 seen in *Xenopus* extracts.[69,72]

(D–F) Crowding can stabilize enzyme-substrate complexes (ESs) relative to the starting reactants. (D) Schematic depiction of a spherical enzyme (E), substrate (S), and complex ES, and the crowder-free volumes surrounding them. (E) A crowding agent, PEG-12,000, promotes complex formation and ATPase activity in the T4 clamp/clamp loader system *in vitro*. The decrease in affinity and rate above 7.5% PEG appeared to be due to crowder-induced aggregation or condensation of the reactants. Data adapted from Jarvis et al.[73] Cartoon adapted from Kelch et al.[74] (F) Crowding-induced increases in the rate constant for a reaction-controlled process as predicted by scaled-particle theory. Here, we have assumed that temperature T = 293°K, r_c = 5 nm, and $r_E = r_S$ = some multiple of r_c as denoted on the *x* axis.

Thus, diffusion slows with increasing cytoplasmic concentration, and the effects are greatest for large species—mesoscale and larger. However, this does not necessarily mean that enzyme-mediated reactions will slow down. It depends upon whether or not they are diffusion controlled.

Scaled-particle theory and the positive effects of crowding on reaction-controlled processes

Despite the effects on diffusion, crowding sometimes makes reactions speed up, a fact well known to biochemists who work with enzymes *in vitro*. One often-cited example is the hydrolysis of ATP *in vitro* by the phage T4 DNA polymerase sliding clamp and clamp loader proteins in the presence of DNA (Figure 2E).[73,74,81] As a crowder (12 kDa polyethylene glycol [PEG]) increases from 0% w/v to 7.5% w/v, the measured dissociation constant decreases ~50-fold and the rate of the process increases ~3-fold. Above 7.5% PEG, the affinity and the reaction rate plummet, apparently due to PEG-induced aggregation of the proteins, but at lower PEG concentrations, there is a decidedly positive effect of crowding on activity.

To see how this positive effect might arise, we return to Equation 3 but now assume that $k_{cat} \ll k_d$. This is termed the reaction limit, and the reaction-controlled rate constant approaches:

$$k_{reaction} = \frac{k_{cat}k_a}{k_d} = \frac{k_{cat}}{K_D}, \qquad \text{(Equation 9)}$$

where K_D is the equilibrium dissociation constant k_d/k_a. *A priori* it is not clear that a crowder would have any effect on k_{cat}, meaning that any effect of crowding on the reaction-controlled rate constant would probably occur through its effect on the equilibrium dissociation constant. The equilibrium constant K_D depends upon k_a and the diffusion coefficients only in so far as the ratio k_d/k_a is affected.

Probably the best-appreciated approach to the question of how the equilibrium constant is affected by crowding comes from Minton,[3,81–83] who assumed that crowders behave like inert

hard spheres and then calculated the entropic cost of arranging those spheres so as to make space for the reactants E and S or for the ES complex (Figure 2D). First let us consider the starting materials, a molecule of E and a molecule of S. We need to produce two crowder-free "holes" to accommodate them in the solution. The holes must actually be bigger than the enzyme or substrate per se. For example, if all species are spherical, the center of a crowder can be no closer than $r_E + r_C$ or $r_S + r_C$ to the enzyme or substrate. Thus, the crowder molecules are excluded from the spherical volumes consisting of an inner kernel of enzyme- or substrate-occupied volume and an outer shell termed excluded volume (Figure 2D).

When an enzyme binds its substrate, some of the enzyme's excluded volume will overlap with the substrate's excluded volume (Figure 2D). It is usually the case that the free energy required to clear the larger single hole (say, a sphere with a radius of $r_{ES} + r_C$) for an ES complex (ΔG_{ES}) will be less than the free energy required to make the two smaller holes for the enzyme and the substrate ($\Delta G_E + \Delta G_S$). For this reason, the crowder stabilizes the complex relative to the reactants, which means that the free energy of complex formation will be more favorable in the presence of crowder. And since

$$\Delta\Delta G = \Delta G_{ES} - \Delta G_E - \Delta G_S, \qquad \text{(Equation 10)}$$

it follows that, in the reaction limit, the effective rate constant increases exponentially as the magnitude of the (negative) $\Delta\Delta G$ value increases:

$$\frac{k_{reaction}}{k^0_{reaction}} = \frac{\frac{k_{cat}}{K_D}}{\frac{k_{cat}}{K^0_D}} = e^{\frac{-\Delta\Delta G}{RT}}. \qquad \text{(Equation 11)}$$

This phenomenon is often referred to as the excluded volume effect.

The quantitative relationship between the crowding and the ΔG values of each of the three species can be estimated theoretically through scaled-particle theory.[3,84] The calculated crowding-induced changes in free energy depend upon the size and shape of the species in question. For spherical species, the expression for the free energy change due to the presence of a crowder at a volume fraction ϕ,[85–87] to third-order in $\frac{\phi}{1-\phi}$, is:

$$\frac{\Delta G_{(S,E,\text{or } ES)}}{RT} = -ln(1-\phi) + (\rho^3 + 3\rho^2 + 3\rho)\frac{\phi}{1-\phi} + (3\rho^3 + 4.5\rho^2)\left(\frac{\phi}{1-\phi}\right)^2 + 3\rho^3\left(\frac{\phi}{1-\phi}\right)^3, \qquad \text{(Equation 12)}$$

where

$$\rho = (r_E, r_S, \text{or } r_{ES}) / r_C.$$

For the example of the T4 DNA polymerase clamp loader and clamp, and for reasonable choices for the parameters that feed into Equation 12 ($r_E = r_S = 4$ nm, $r_{ES} = 5$ nm, $r_C = 1.68$ nm), the prediction is that the affinity of the clamp loader for the clamp and the rate of ATP hydrolysis should be ~5-fold higher in 7.5% PEG than in uncrowded buffers. As mentioned above, experimentally the affinity ($1/K_D$) was found to increase by about ~50-fold and the rate was found to increase by ~3-fold. Thus, the theory qualitatively accounts for the positive effects of crowding, but there are some discrepancies in the quantitative details.

Like the negative effects, the positive effects of crowding are expected to vary depending upon how big the enzyme and substrate molecules are compared with the crowders. This prediction again comes from the statistical thermodynamics of hard sphere liquids (Equation 12). If the enzyme and substrate are smaller than the crowder, the positive effects of crowding are minimal. However, once the size of the enzyme and substrate exceeds the size of the crowder, the rate enhancement explodes (Figure 2F). Thus, for reaction-controlled processes, it is large enzymes and substrates that are expected to be most positively affected by crowding, just as for diffusion-controlled processes, it is large enzymes and substrates that are expected to be most negatively affected by crowding.

Note that although here we have focused on the positive effects of crowding on enzyme-mediated reaction rates, one could equally well apply this approach to other important biological processes, such as complex formation, protein folding, and macromolecular condensation.[3,4,6]

A combined expression for $k_{effective}$ when the system is neither at the reaction limit nor the diffusion limit

The above sections consider reactions at the extremes, the diffusion and reaction limits. We can combine the positive and negative effects of crowding into a single equation for a far-from-saturation system operating anywhere between the diffusion limit and the reaction limit. From the definitions:

$$k_{effective} = \frac{k_{cat}k_a}{k_{cat}+k_d}, k_{diffusion} = k_a, \text{and } k_{reaction} = \frac{k_{cat}}{K_D} = \frac{k_{cat}k_a}{k_d}, \qquad \text{(Equation 13)}$$

it follows that

$$k_{effective} = \frac{k_{reaction}k_{diffusion}}{k_{reaction}+k_{diffusion}}. \qquad \text{(Equation 14)}$$

Substituting Equations 8 and 11 into Equation 14 yields

$$k_{effective} = \frac{k^0_{reaction}e^{\frac{-\Delta\Delta G}{RT}}k^0_{diffusion}e^{-g\phi}}{k^0_{reaction}e^{\frac{-\Delta\Delta G}{RT}}+k^0_{diffusion}e^{-g\phi}}, \qquad \text{(Equation 15)}$$

which can be simplified to

$$k_{effective} = \frac{k^0_{reaction}k^0_{diffusion}}{k^0_{reaction}e^{g\phi}+k^0_{diffusion}e^{\frac{-\Delta\Delta G}{RT}}}. \qquad \text{(Equation 16)}$$

In terms of the more usual rate constants k^0_a, k^0_d, and k^0_{cat}, the equation becomes

$$k_{effective} = \frac{k^0_a k^0_{cat}}{k^0_{cat}e^{g\phi}+k^0_d e^{\frac{-\Delta\Delta G}{RT}}}. \qquad \text{(Equation 17)}$$

We can get a feel for Equation 17 by plotting it. Figure 3A shows the effective rate constant as a function of the crowding ϕ for a system that is midway between the reaction-controlled and diffusion-controlled limits. The predicted dependence is biphasic—there is a small increase in the effective rate constant

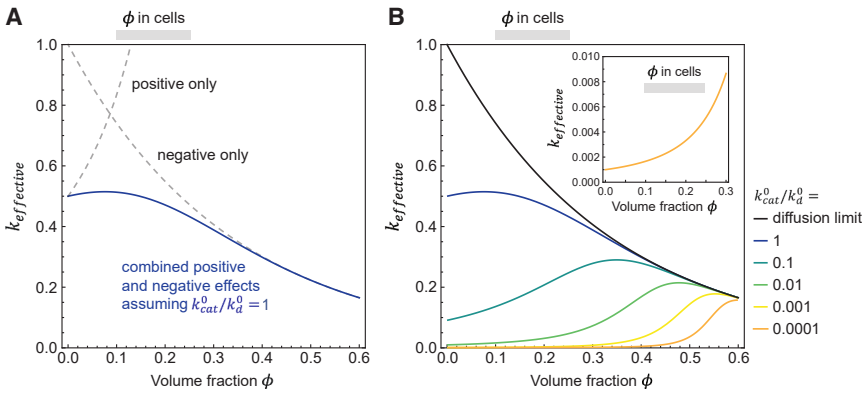

Figure 3. How the positive and negative effects of crowding combine for processes that lie between the reaction limit and the diffusion limit

(A) The theoretical effective second-order rate constant ($k_{effective}$) for an unsaturated enzyme-mediated reaction, assuming that the process is neither reaction controlled nor diffusion controlled in buffer, and $k^0_{cat} = k^0_d$. For comparison, the dashed lines show the expected positive and negative effects of crowding individually.

(B) The theoretical effective second-order rate constant at other ratios of k^0_{cat} to k^0_d. Note that the "average" enzyme has a k^0_d value ~10,000× larger than its k^0_{cat} value, which corresponds to the orange curve. The inset shows the orange curve $\left(\frac{k^0_{cat}}{k^0_d} = 0.001\right)$ with a zoomed-in *y* axis scale so that the changes in $k_{effective}$ over the range of φ values expected in cells are easier to appreciate. For both panels, parameters are $r_E = r_S = r_C = 5$ nm, $g = 3$, $k^0_{diffusion} = 1$, and $k^0_{reaction}$ = various values. Note that $k^0_{reaction}/k^0_{diffusion} = k^0_{cat}/k^0_d$.

at small values of ϕ and then a more dramatic decrease at higher values. For these parameter choices, the positive effects of crowding are fairly minimal. However, as the assumed value of k^0_{cat} decreases and the process gets closer to the reaction limit, the positive effects of crowding become more dominant (Figure 3B). Since most enzymes are thought to be fairly inefficient, with a typical enzyme estimated to have 10,000 non-productive collisions for every reaction,[88] the expectation is that the positive effects of crowding should dominate the picture (Figure 3B, orange curve) until the crowding becomes very high ($\phi \approx 0.6$, near glass transition and the random packing density of spheres), if this theoretical framework and these assumed parameters are reasonable.

Experimental data on the effects of crowding on reaction rates

Having considered what might be expected for the effects of crowders, we turn to what does happen. The effects of crowding on reaction rate constants can be relatively easily assessed through traditional *in vitro* experiments. Ideally, a crowder in such experiments should mimic the cell's natural crowders in terms of charge, size, and interactions with other macromolecules. In practice, protein-sized uncharged polymers like dextran or PEG are probably the most frequently used crowders for *in vitro* work, although purified ribosomes[76] and freeze-dried *E. coli* extracts[89,90] have also been used. Data from two such *in vitro* experiments are shown in Figures 4A and 4B. T7 RNA polymerase-catalyzed transcription[91] (Figure 4A) and mitogen-activated protein kinase (MAPK) signaling[92] (Figure 4B) are both affected by artificial crowders in a biphasic fashion. Their apparent second-order rate constants first increase with crowding and then decrease at higher crowding levels, showing both the positive and negative effects expected theoretically (Figures 4A and 4B). These results together with other *in vitro* studies[93–95] support the idea that crowding could have significant impacts on biochemical reactions in living cells, assuming these artificial crowders faithfully mimic cellular crowders.

But, as they say, *in vivo veritas* (and perhaps *in extracto veritas* as well, given that *Xenopus* extracts are undiluted, possess "natural" crowders, and are remarkably functional[17,18,98–104]), and recent work has yielded some estimates of reaction rate constants as a function of crowding in *Xenopus* egg extracts[68,71] (Figures 4C–4I) and in yeast cells[75,96,97] (Figures 4J and 4K). In the egg extracts, where the cytoplasmic protein concentration is on the order of 60 mg mL^{-1}, crowding was manipulated by diluting 1× extracts with either buffer or a protein-free cytosolic filtrate and by concentrating the extracts to 2× with a centrifugal filter and then diluting back to the desired concentration. In principle, these experiments perturb the macromolecule concentration—enzymes, macromolecular substrates, and endogenous crowders—without affecting the concentrations of ions or low molecular weight biomolecules. A great variety of assays can be utilized in this cell-free system, provided the assay is sufficiently specific for the activity in question. For the yeast, where the normal cytoplasmic protein concentration is probably greater than 100 mg mL^{-1},[105,106] crowding was varied by adjusting the external osmolarity, which perturbs the concentrations of all intracellular species. With this approach, assays are limited to cell-permeant molecules and genetically encoded probes. For context, recall that the protein concentration among different cell types can vary by as much as 5-fold (e.g., 60 mg mL^{-1} for *Xenopus* eggs vs. 300 mg mL^{-1} for human erythrocytes), so the differences in crowding achieved here experimentally could be informative.

Nine processes have been examined, ranging from the cleavage of small fluorogenic peptides by caspases to protein synthesis by mesoscale ribosomes, mRNAs, and polysomes (Figures 4C–4K). *A priori*, it seems plausible that in addition to the positive and negative effects of crowding described above, there could be critical points in the activity vs. crowding curves from condensation effects.[107,108] For example, at high crowder concentrations, the enzyme or substrate could be driven into an inactive state in a condensate, or at low crowder concentrations, they could be driven out of an active condensed reaction cauldron. However, this was not seen for any of these nine processes. In all cases, the values of the rate constants varied smoothly with the crowding (Figures 4C–4K).

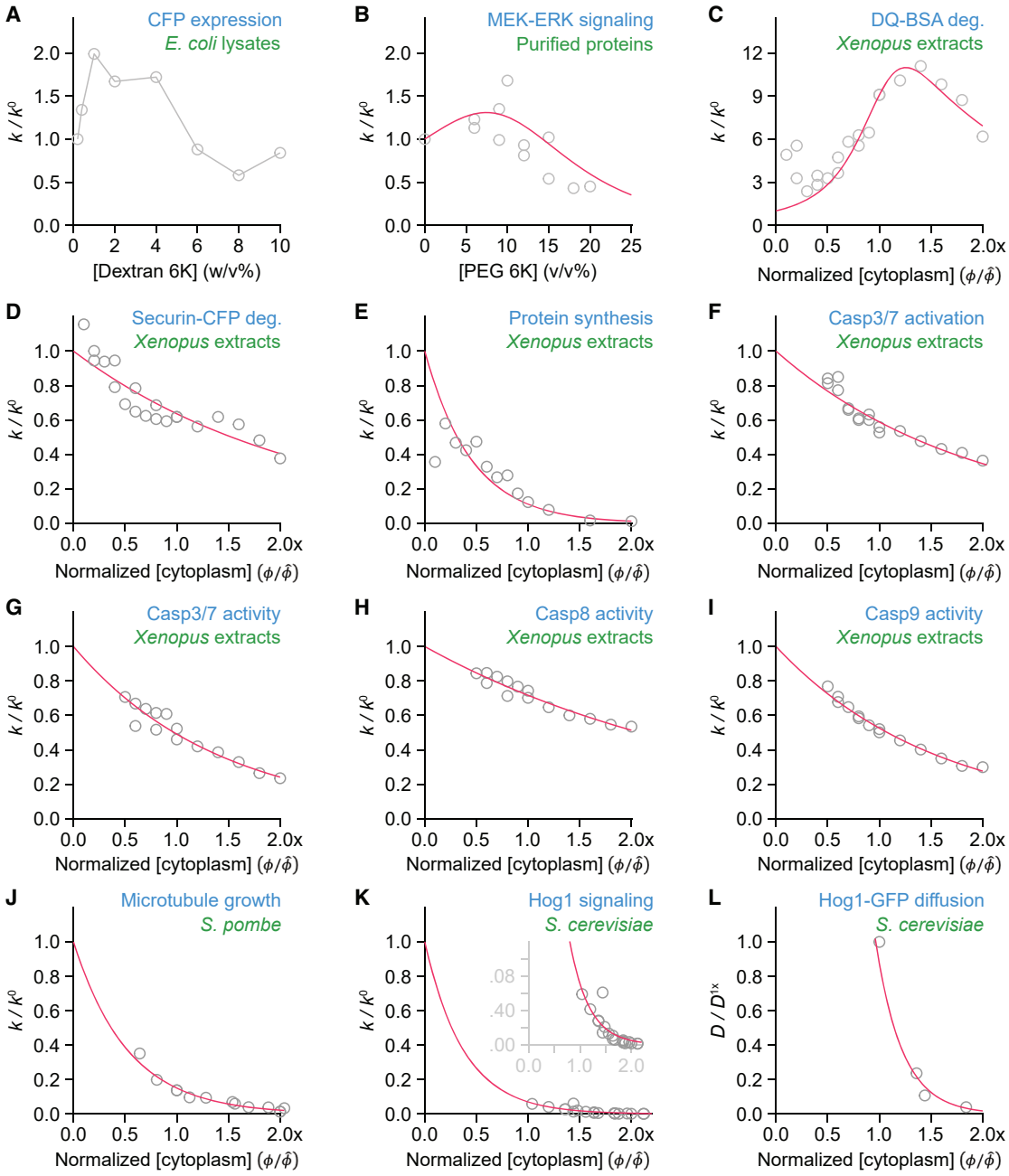

Figure 4. *In vivo*, cytoplasmic crowding generally decreases the effective second-order rate constant *k*

(A and B) The effects of crowding agents on T7 RNA polymerase activity (A) and MEK-ERK signaling (B) *in vitro*.

(C–I) The effects of cytoplasmic concentration on various enzyme-mediated processes in *Xenopus* egg extracts.

(J–L) The effects of cytoplasmic concentration on microtubule growth (J) and Hog1 signaling (K) and diffusion (L) in yeasts.

The cytoplasmic concentration (C) is expressed relative to the cytoplasmic concentration of undiluted extracts or unperturbed yeasts $(\hat{\phi})$. In (B) and (C), the red curves are fits of Equation 17 to the data. In (D–L), Phillies's law (Equation 8) is fitted to the data. (A) Tan et al.[91] (B) Aoki et al.[92] (C–E) Chen et al.[68] (F) Huang et al.[71] (G–I) The present work. (J) Molines et al.[96] (K and L) Miermont et al.[75] and Babazadeh et al.[97] The fit in (C) used scaled-particle theory for spherocylinders as described in Zhou et al.[3], assuming that both the crowder and BSA are spheres of 5 nm radius, that the proteasome is a spherocylinder with 20 nm cylindrical length and spherical caps of 5 nm radius, that the BSA-proteasome complex is also a spherocylinder with 5 nm-radius spherical caps but with 26.7 nm cylindrical length, and, finally, that $\frac{k^0_{diffusion}}{k^0_{reaction}} = 30$ (or $\frac{k^0_{cat}}{k^0_d} = 0.03$) and $g = 3$. Activity of caspases (G–I) was measured in fully apoptotic *Xenopus* extracts by adding fluorogenic substrates (Z-Asp-Glu-Val-Asp)$_2$-R110 (G), (Ac-Ile-Glu-Thr-Asp)$_2$-R110 (H), and (Z-Leu-Glu-His-Asp)$_2$-R110 (I) and then promptly recording the fluorescence increase. The apoptotic extracts were freshly prepared by treating live extracts with 2 μM horse cytochrome c for 30 min. Details of the experimental procedure are described in Huang et al.[71] Each data point is the mean from 3 biological repeats. (J–L) We estimated the yeast (cytoplasm) from the reported cell volume *V*

(legend continued on next page)

One of the reactions examined, proteasome-mediated degradation of a dye-quenched BSA protein (DQ-BSA) (Figure 4C), showed the biphasic dependence expected from our theoretical framework (Equations 16 and 17). The inferred apparent rate constant increased with the cytoplasmic concentration to a maximum at approximately 1.3× the concentration of undiluted *Xenopus* extracts. The magnitude of the increase in rate constant was substantial, about 10-fold based on the ratio of the peak value to the value extrapolated to zero crowding. Above a concentration of 1.3×, the rate constant decreased. Overall the experimental data could be fitted fairly well by Equation 17, with some disagreement at low cytoplasmic concentrations.

However, in all of the other examples (Figures 4D–4K), there was no obvious positive effect of crowding on the measured rate constants. Instead, there was an exponential decrease in rate constant with increasing cytoplasmic concentration, as expected from Phillies's law (Equation 8) and from Equations 16 and 17 in the diffusion limit. This monotonic decrease was found for five processes with proteins or macromolecular complexes as the relevant enzymes and substrates and for the three processes where the substrates were small molecules (Figures 4D–4I). Note that the negative effect of crowding on microtubule growth in cells[96] (Figure 4E) was the opposite of the effect of artificial crowders on microtubule growth *in vitro*.[94] The steepness of the decrease in rate should be determined by the parameter g, which describes how sensitive the diffusion coefficient is to changes in crowding (Equation 7). For six of the processes, including all but one of the reactions measured in *Xenopus* egg extracts, the value of g is fairly close to 3 (range 1.3 to 3), which is about what would be expected for a diffusion-controlled process involving small molecules or protein-sized species. For the remaining three processes, it was substantially larger (range 7.6 to 16), more consistent with reactions involving mesoscale species. These included translation in *Xenopus* egg extracts ($g = 8.8$), which does involve mesoscale species (polysomes, mRNAs, and ribosomes). The other two processes were the two assessed *in vivo* in fungi: microtubule growth and the activation of Hog1, a MAP kinase that responds to increases in osmolarity via a system that includes the putative osmosensors Sln1, Msb2, and Hkr1.[109] Hog1's scaling factors for diffusion ($g = 15.5$) and activation ($g = 16.0$) were similarly high, suggesting that either the interacting species are larger than previously appreciated or that yeast cytoplasm is more crowded than *Xenopus* egg cytoplasm.

From these examples, it appears that crowding often decreases reaction rate constants in cells and extracts, and if it has much of a positive effect, most of the processes examined so far do not show it.

Are all of these systems diffusion controlled?

As mentioned above, surveys of enzyme kinetic data obtained *in vitro* show that most enzymes operate well below the diffusion limit.[88,110] Typical values of k^0_{cat}/K^0_d are $\sim 10^5$ M^{-1} s^{-1}, whereas from the Smoluchowski equation, k^0_a is $\sim 10^9$ M^{-1} s^{-1}. One might think then that the negative effects of crowding should be overwhelmed by the positive effects, as shown in Figure 3.

Yet none of the results from the *Xenopus* extract or yeast systems (Figure 4) showed a monotonic increase in the effective second-order rate constant out to very high values of ϕ as this analysis would predict. In the one case where the rate constant at least initially increased with increased crowding, DQ-BSA proteolysis (Figure 4C), it decreased above a cytoplasmic concentration of 1.3×, which is about what would be expected if $k^0_{cat}/k^0_d \approx 0.1$ (Figure 3B, blue-green curve), three orders of magnitude closer to the diffusion limit than the average enzyme is reported to be.[88,110] The remaining reactions behaved as though they were even closer to the diffusion limit. Over the ranges of cytoplasmic concentrations examined, the rate constant vs. ϕ curves were indistinguishable from those expected for a diffusion-controlled reaction. This is a conundrum.

One possible explanation is that the reactions are not diffusion controlled, as it appears, but the increases in cytoplasmic concentration slowed both translational diffusion and the intra-complex motions that are required for catalysis. If so, there could be a decrease in the k_{cat} value that outweighs the decreased K_D value predicted by scaled-particle theory, making $k_{reaction} = \frac{k_{cat}}{K_D}$ decrease with increased crowding.

A second possibility is again that the reactions are not diffusion controlled, but that crowding promotes not only the formation of ESs but also the formation of unproductive complexes, such as enzyme aggregates, substrate aggregates, or off-target combinations of enzymes with wrong substrates or substrates with wrong enzymes.[111] Indeed, in the *in vitro* experiments shown in Figure 2E, it was noted that at the PEG-12,000 concentrations where the reaction rate fell, a visible precipitate was evident.[72] Either or both of these possibilities could allow a reaction-controlled process to overall be negatively affected by crowding.

A third possibility is that the positive effects of crowding on the equilibrium between the reactants and the ES complex might be smaller than expected from scaled-particle theory or might even be opposite to what scaled-particle theory predicts because of attraction between the crowder and the reactants. Scaled-particle theory assumes that the reactants are neither attracted to nor repulsed by the crowders, so the entropic gains from the excluded volume effect outweigh any enthalpic penalties due to losing favorable interactions between the crowder and the enzyme and/or substrate. When this assumption is violated, as likely is *in vivo*,[70,112–115] the positive effects of crowding can be

under osmotic shock[75,97,96] and assuming [cytoplasm]$\propto \frac{1}{V}$. (J) Microtubule growth rates were converted from μm min^{-1} to nM min^{-1}, knowing that there are 13 tubulin dimers per turn on a microtubule and that each tubulin dimer is 8 nm long. We assumed that the number of plus (growing) ends remained constant. (K and L) The characteristic time for Hog1 phosphorylation $\tau_{phosphorylation}$ or translocation $\tau_{translocation}$ were converted to first-order rate constants by $k = \frac{1}{\tau_{phosphorylation}}$ or $k = \frac{1}{\tau_{translocation}}$. The apparent second-order rate constants were obtained by dividing the first-order rate constant by [cytoplasm]. Diffusion coefficients D as reported in Miermont et al.[75] were calculated from the residence time τ_{FCS} measured by fluorescence correlation spectroscopy using the relation $D \propto \frac{1}{\tau_{FCS}}$ in Babazadeh et al.[97]

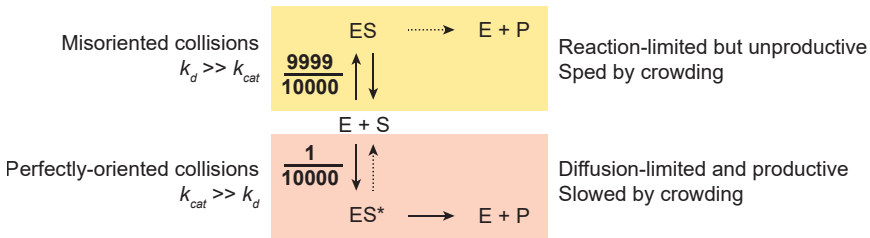

Figure 5. A simple scheme to show how a reaction might be diffusion controlled and its rate might be affected by changes in diffusion rates even though only 1 in 10,000 reactions is productive

diminished or even reversed.[5,6,116] This could explain the overall negative effects of crowding on rate constants.

These three hypotheses all assume that the processes examined are reaction controlled. A fourth possibility is that the reactions are in fact diffusion controlled (and hence are affected by things that slow diffusion), even if they are operating well below the Smoluchowski limit ($\sim 10^9\ M^{-1}\ s^{-1}$) and have k_d values that are much larger than their k_{cat} values. If we view the binding of an enzyme to a substrate as arising from a collision between two non-attracting, non-repelling hard spheres, then the vast majority of collisions will be non-productive because the substrate does not hit the enzyme's active site or does hit it but is oriented incorrectly, whereas the collision rate calculation implicitly assumes that 100% of the reactants' surfaces are potentially reactive and all orientations are equivalent.[117,118] If the reactive target area of an enzyme is, say, 3% of its surface area, and the target area on a similarly sized substrate is also 3% of its surface area, then only 0.1% of collisions will be between the two target areas. If only 10% of the target-to-target collisions are oriented correctly, then the maximum reaction rate would be 10,000-fold lower than the Smoluchowski limit.[118] Perhaps the small number of enzymes that do operate near the Smoluchowski limit (e.g., carbonic anhydrase, fumarase, and superoxide dismutase) achieve this by short-range forces that allow the colliding enzyme and substrate's active sites to find each other through attractive interactions no matter how misoriented they start out.[119] Perhaps many "average" enzymes are acting more like hard spheres and so can be diffusion controlled well below the Smoluchowski limit.

One way to think about this possibility is shown schematically in Figure 5. We imagine that 9,999 out of 10,000 collisions will be oriented incorrectly and will result in the transient formation of a non-productive ES complex. For this ES complex, the k_d value is large and k_{cat} is essentially zero, so the process is reaction controlled, but it contributes almost nothing to the overall rate of product formation. One out of 10,000 collisions will be oriented correctly, producing a complex termed ES*. We imagine this correctly oriented complex to never dissociate back to reactants and to always go on to form product. This process would therefore be diffusion controlled, and decreasing k_a by decreasing the diffusion coefficients of the enzyme and substrate would decrease the rate of the process in accordance with Phillies's law. Thus, even though the system overall is operating far from catalytic perfection, product formation is diffusion controlled.

If this is true, then the extent to which reaction rates change with crowding should match quantitatively with the scaling of diffusion. This is in fact the case for Hog1 signaling in yeast. The g values for Hog1 signaling and diffusion are essentially equal (16.0 vs. 15.5) (Figures 4K and 4L). Likewise, for six of the processes assessed in *Xenopus* extracts, the g values for the reactions are reasonably close to 3, the value measured for diffusion of proteins and small molecules (Figures 4C, 4D, and 4F–4I). This hypothesis could be tested further by comparing one substrate to a second one that has similar collision and dissociation kinetics but a smaller k_{cat} value *in vitro*. If the scheme proposed in Figure 5 pertains, the expectation is that the low k_{cat} substrate should not exhibit slow reaction rates *in vivo* or in extracts.

Maximal rates rather than maximal rate constants?

So far, we have focused on the effective second-order rate constants for the reaction, but the concentrations of enzyme and substrate also figure into the overall rate, and these concentrations are also expected to change as the cytoplasm is concentrated or diluted.[120] To gauge how crowding would be expected to affect reaction rates, we need to take this into account.

If we assume simple second-order kinetics (Equation 3) and assume that as we dilute or concentrate cytoplasmic components in a cell or in an extract, the concentrations of the enzyme and the substrate change proportionally, then:

$$rate = k_{effective}[E][S] = k_{effective}\left(\frac{\phi}{\widehat{\phi}}\right)^2[\widehat{E}][\widehat{S}], \qquad \text{(Equation 18)}$$

where $\widehat{\phi}$ is the nominal 1× cytoplasmic concentration (e.g., $\widehat{\phi} \approx 0.25$), and $\widehat{E}$ and $\widehat{S}$ are the concentrations of the enzyme and substrate at this cytoplasmic concentration. Assuming that the rate is diffusion controlled, we can substitute a Phillies's law expression (Equation 8) for $k_{effective}$:

$$rate = k^0_{diffusion}e^{-g\phi}\left(\frac{\phi}{\widehat{\phi}}\right)^2[\widehat{E}][\widehat{S}]. \qquad \text{(Equation 19)}$$

More simply:

$$rate = Ae^{-g\phi}\left(\frac{\phi}{\widehat{\phi}}\right)^2. \qquad \text{(Equation 20)}$$

In this scheme, the negative effects of crowding on the rate constant contribute to the decreasing exponential factor, and the positive effects of crowding on the concentrations give us the increasing ϕ^2 factor. Note that a single parameter, g, determines the shape of the rate vs. crowding curve, and the proportionality factor A stretches the curve up or down. We can also define a crowding sensitivity factor χ to equal $\frac{drate}{d\phi}$ at $\phi = \widehat{\phi}$. If χ is positive, the reaction rate increases when the system is perturbed upward from a 1× cytoplasmic concentration; if χ is negative, it decreases; and if χ is zero, the curve is flat.

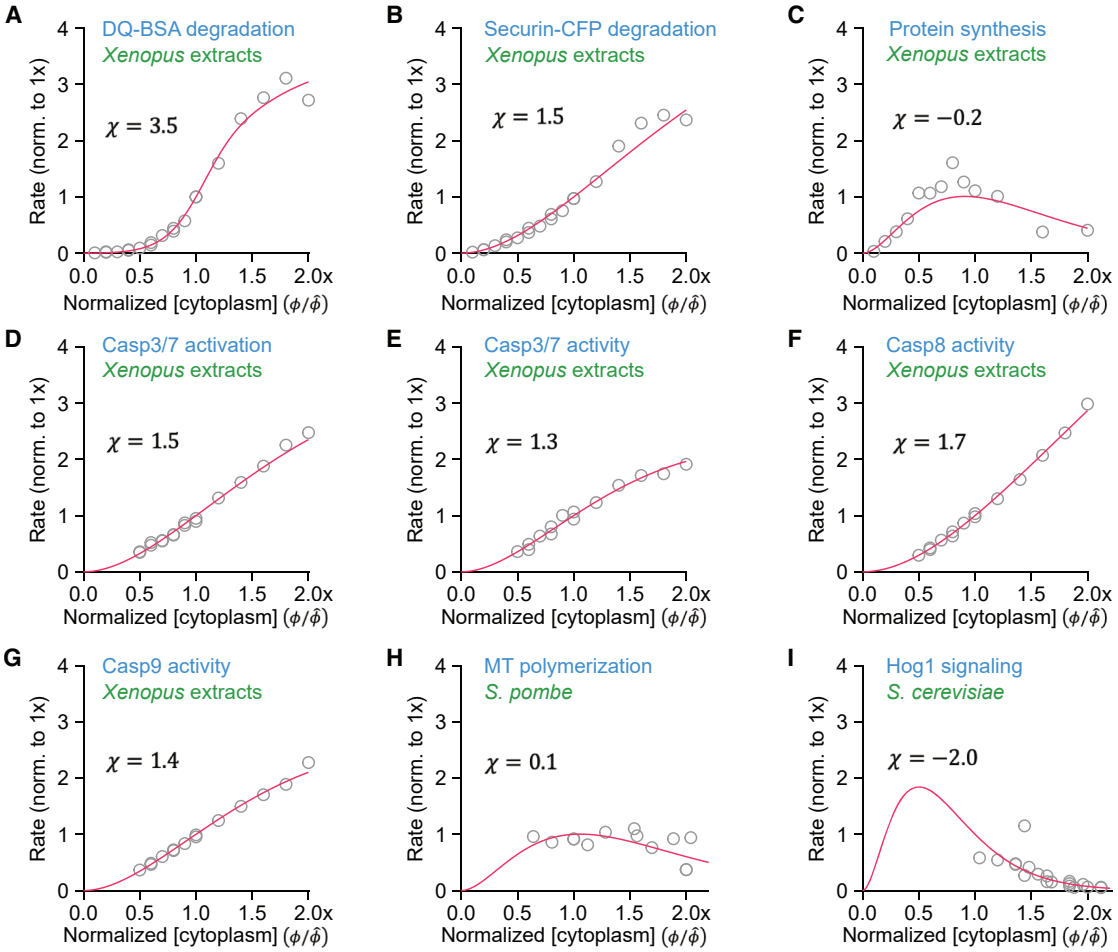

Figure 6. The overall rates of various processes in extracts or *in vivo* as a function of the cytoplasmic concentration

Here, we have assumed that $rate = k[E][S]$, that k varies with the cytoplasmic concentration as shown in Figure 4, and that $[E]$ and $[S]$ increase linearly with the cytoplasmic concentration ϕ. The data here were calculated by taking the rate constant data from Figure 4 and multiplying by $(\phi/\hat{\phi})^2$, where the normalization factor $\hat{\phi}$ represents the cytoplasmic concentration of an undiluted extract or an unperturbed yeast cell. The red curves are fits of Equation 20 $\left(rate = Ae^{-g\phi}\left(\frac{\phi}{\hat{\phi}}\right)^2\right)$ to the data. The y axes are scaled so that the fitted curves equal 1 at a cytoplasmic concentration of 1×.

(A–C) Chen et al.[68]
(D) Huang et al.[71]
(E–G) The present work.
(H) Molines et al.[96]
(I) Miermont et al.[75] and Babazadeh et al.[97] The quantity χ is the slope of the fitted curve at a cytoplasmic concentration of 1×.

Figure 6 shows fits of Equation 20 to the experimental rate data. For six of the processes—DQ-BSA degradation (Figure 6A), securin degradation (Figure 6B), caspase 3/7 activation (Figure 6D), caspase 3/7 activity (Figure 6E), caspase 8 activity (Figure 6F), and caspase 9 activity (Figure 6G)—the rate of the process increases monotonically with the concentration of the cytoplasm. These are the six processes with the lowest g values (g = 1.3 to 3) and relatively low molecular weights for the enzymes and reactants. For two of the processes, protein synthesis in *Xenopus* extracts (Figure 6C) and Hog1 activation in budding yeast (Figure 6I), with larger g values indicating that reactants behave more like mesoscale proteins, the response is biphasic, and the rate of the process decreases when the cytoplasmic concentration rises above 1×. In the case of protein synthesis and degradation (Figures 6A–6C), this makes for a homeostatic mechanism: if the protein concentration increases, the rate of synthesis goes down and the rate of degradation goes up, driving the system back toward its original protein concentration.[68] Stability and homeostasis seem like such fundamental and commonplace aspects of living systems that it is likely that evolution has arrived at a variety of strategies to ensure them. That said, the simplicity of this differential scaling mechanism suggests that it is probably not unique to protein concentration homeostasis in frog eggs.

For one process, microtubule growth in yeast, the increase in rate due to increasing the concentration of the reactants almost

perfectly balances the decrease in rate constant due to Phillies's law effects, making the rate of microtubule growth nearly invariant (Figure 6H). A similar invariance has been demonstrated for the speeds of mitotic and apoptotic trigger waves in *Xenopus* extracts, and again the robustness appears to arise from the nearly precise cancelation of opposing positive and negative factors.[71] Irrespective of these interesting details, overall the positive effect of increasing the reactant concentrations appears to dwarf any positive effect from reaction-controlled kinetics and thermodynamic stabilization of the ES complex, at least for these particular processes.

What drives hyperosmolarity-induced condensation?

Some of the recent interest in the effects of cytoplasmic crowding has arisen out of the study of biomolecular condensates, and we will conclude here with a discussion of how the principles and experimental results reviewed here bear on these fascinating processes. Hyperosmotic shock shrinks cell volume and drives many proteins into condensates.[121] These condensates could bring the reactants together and facilitate their reactions, or they could halt reactions by demixing the reactants into separate phases. Condensates form abruptly when the environment shifts, reflecting the criticality of condensation. For example, condensation is thought to be a switch that turns on and off the cell's dedicated osmosensor proteins, which then relay the information to the cell's volume homeostat.[122–126] With many proteins besides the canonical osmosensors (e.g., *Arabidopsis* DCP5 and mammalian Wnk1 and Ask3[124–126]) also forming condensates, the cell may sense and respond to osmotic shocks in a distributed manner,[77] and it may buffer the availability of exchangeable water—perhaps up to 5% of the cell volume—by releasing the water molecules bound to protein surfaces when those proteins undergo condensation.[127]

As the cell volume shrinks, reactant concentrations and crowding go up. In theory, both could drive the system through the critical point and lead to condensation. Yet, in light of the data presented in Figures 4 and 6, where excluded volume effects have mostly taken a backseat, it may be of interest to quantify the respective contributions from excluded volume (positive) effects or concentration (positive) effects and to what extent viscosity and diffusion (negative) effects also contribute. This could tell us what the cell is actually sensing as it tries to tune its volume to the biomass that it has accrued: is it concentration, or is it crowding? And, although not guaranteed, we may find different categories of enzymes: some that are diffusion controlled, meaning they are as efficient as they can be given the physicochemical nature of the cell interior, and others that are less efficient but can harness the positive effects of crowding as a means of regulation.

Conclusions

Here, we have reviewed two ways that cytoplasmic crowding might be expected to affect biochemical reaction rates. For reaction-controlled processes, cytoplasmic crowding is expected to speed the process via thermodynamic stabilization of the ES, and the magnitude of this expected effect can be estimated through scaled-particle theory. For diffusion-controlled processes, cytoplasmic crowding is expected to slow the reaction by slowing diffusion, with the rate constant decreasing exponentially with increased crowding. There are now a handful of reports on how crowding affects various processes—protein synthesis, protein degradation, caspase activation and activity, Hog1 activation, and microtubule growth—*in vivo* or in *Xenopus* egg extracts. While one reaction, BSA degradation in extracts, shows some positive effect of crowding on the effective second-order rate constant for the process, the other eight do not. This suggests, first, that reactions *in vivo* may be diffusion controlled more often than one might expect from comparing *in vitro* kinetic data to the Smoluchowski limit and, second, that the positive effects of cytoplasmic concentration on reaction rates mainly arise from the changes in reactant concentration rather than thermodynamic stabilization of ESs.

That said, the data are just now arriving, so as much as anything, this discussion is a call to the community to see how your favorite aspect of cellular biochemistry changes when you make your favorite cell more or less crowded. In addition to the direct approaches used in the studies examined here, one can also force polysomes to disassemble in intact cells, which has the effect of decreasing cytoplasmic crowding,[128,129] and there will likely be other approaches developed as well. Our hope is that through these approaches we might arrive at a much more comprehensive understanding of how the crowded environment of the cell shapes the biochemistry of life.

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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